



PROJECT EXPO-2026

IoT BASED SMART TROLLEY USING RFID TAGS, FIREBASE DATABASE AND BLE NAVIGATION

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PROBLEM STATEMENT

The Problems faced by the customers while shopping groceries

- Manual billing causes long queues
 - Customers struggle to locate products
 - Need for automation and navigation in retail
- Objective: Develop a trolley that automates billing and guides users using BLE signals

PROPOSED SOLUTION



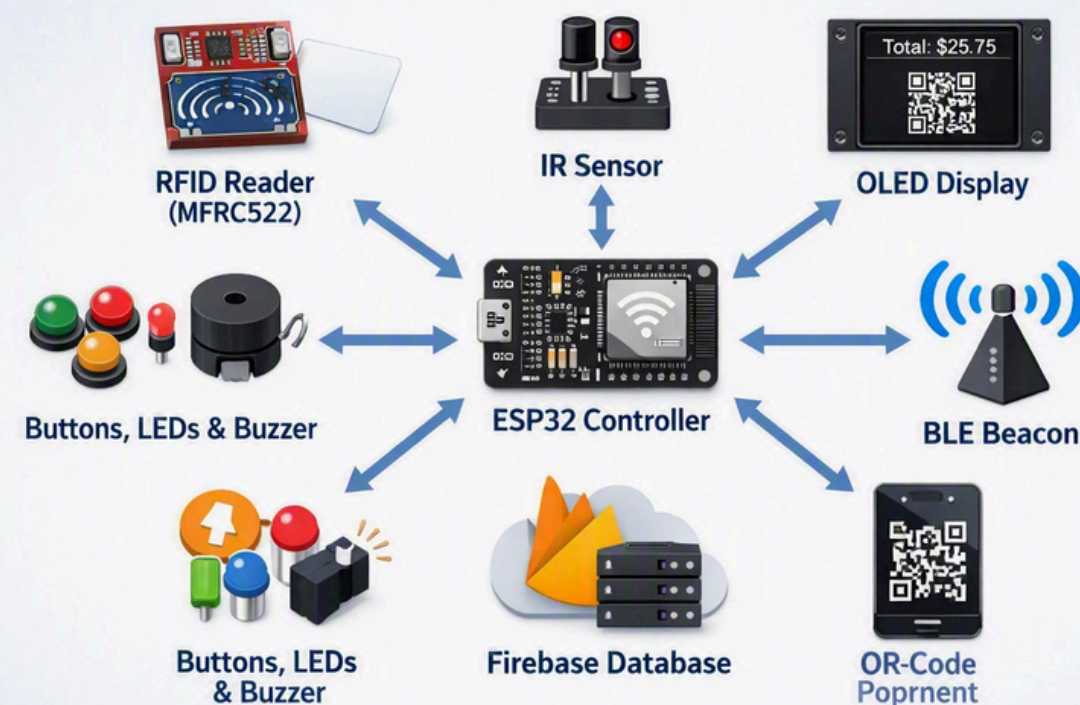
Key Advantages for our solution

- RFID-based automatic billing
- BLE beacon navigation system
- Display real-time updates on OLED
- Buzzer and LEDs for item/payment indication
- ESP32 DevKit as the central controller
- Enable QR-based payment



System Components

Made with AI



SYSTEM ARCHITECTURE

COMPONENTS

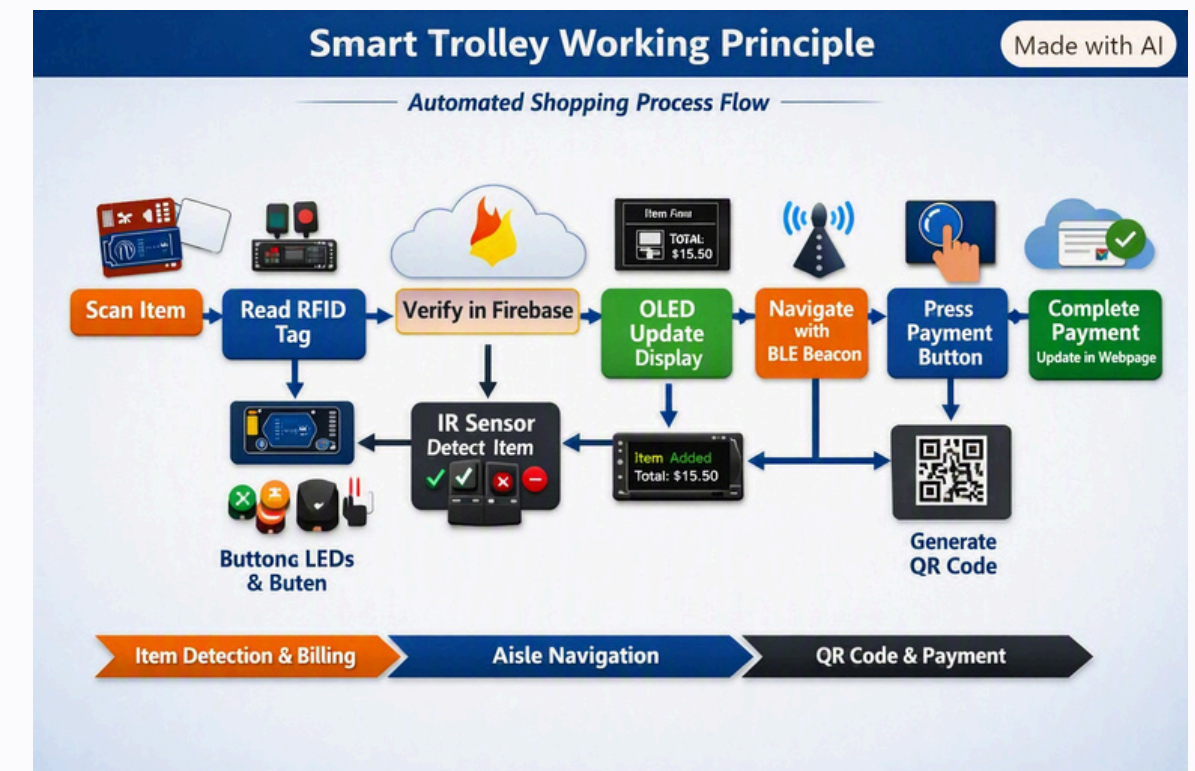
- **ESP32 microcontroller**
- **RFID module (MFRC522)**
- **IR sensor**
- **OLED display**
- **BLE beacons**
- **Firebase database**
- **buzzer, LEDs**
- **push buttons (checkout and navigation button)**



WORKING PRINCIPLE

Step-by-step process of how our smart trolley operates

- Firebase Database: Stores item details linked with unique RFID tags.
- RFID Module: Scans the tag and identifies the item from Firebase.
- IR Sensor: Detects whether the item is added or removed from the trolley.
- OLED Display: Shows item name, price, total bill, and navigation status.
- BLE Beacon Navigation: Each aisle has a beacon transmitting Bluetooth signals; ESP32 receives and calculates RSSI for navigation.
- Navigation Control: Starts and stops using a dedicated button.
- Payment Process: Pressing the payment button generates a QR code on OLED.
- Firebase Update: After payment, shopping details are updated on the webpage.



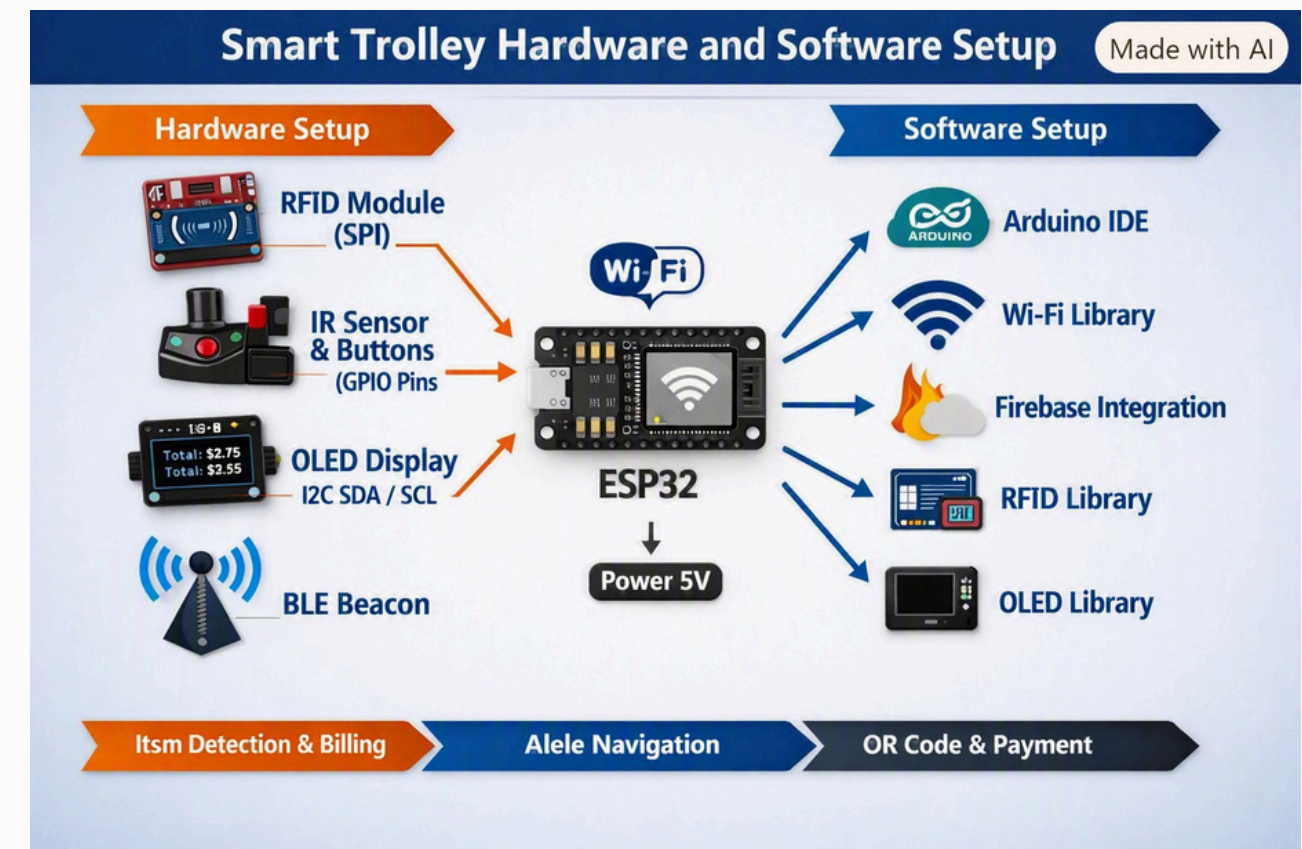
HARDWARE AND SOFTWARE SETUP

Hardware Setup

- RFID connected via SPI
- OLED via I2C (SDA, SCL)
- IR sensor, buttons, buzzer, LEDs via GPIO
- BLE beacons placed in aisles
- ESP32 powered by 5V supply

Software Setup

- Arduino IDE
- Libraries: WiFi.h, FirebaseESP32.h, Adafruit_SSD1306.h, MFRC522.h, NimBLE.h
- Firebase project configuration
- ESP32 code upload



RESULTS AND ADVANTAGES

- **Results:**
 - Real-time billing on OLED
 - Accurate item detection
 - Indoor navigation via BLE RSSI
 - QR code generation for payment
 - Webpage update after checkout
- **Advantages:**
 - Faster checkout
 - Smart navigation
 - Accurate billing
 - Reduced manpower
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FUTURE SCOPE & CONCLUSION

Future Scope:

- Multi-beacon triangulation for precise indoor positioning
- Cloud-based billing and mobile app integration

Conclusion:

- Smart Trolley combines IoT and embedded systems to make shopping smarter, faster, and more efficient



Thank You !

